CY34/CI34

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RAMAIAH
Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END EXAMINATIONS – APRIL 2023

Program	:	**B.E. - CSE (Cyber Security) / CSE (Artificial Intelligence and Machine Learning)**	**Semester**	:	**III**
Course Name	:	**Computer Organization and Architecture**	**Max. Marks**	:	**100**
Course Code	:	**CY34 / CI34**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1.	a)	Simplify the following three variable expression using Boolean algebra Y=$\sum m(0,2,6,7)$.	CO1	(04)
	b)	Design half adder and Full adder circuit.	CO1	(06)
	c)	Briefly describe the working of clocked RS flip-flop with circuit diagram and truth table.	CO1	(05)
	d)	Describe the working of 4 bit asynchronous counter with a neat diagram.	CO1	(05)
2.	a)	Simplify the following three variable expression using Boolean algebra Y=$\sum m(1,5,7)$ and Y=пM(0,2,6,7).	CO1	(06)
	b)	With neat circuit diagram describe half Subtractor and Full Subtractor.	CO1	(06)
	c)	Define race condition? With the help of Truth table Explain JK flip-flop.	CO1	(08)

UNIT - II

3.	a)	Perform sequential multiplication on (10*15) using version-1 multiplication algorithm.	CO2	(06)
	b)	Describe any two types of MIPS instruction formats with an example for each type.	CO2	(06)
	c)	With an example, explain the different ways in which the operands can be specified in an assembly language instruction.	CO2	(08)
4.	a)	Perform binary division on the following using improved division algorithm. Dividend=15 and Divisor=4.	CO2	(06)
	b)	Describe the two types of MIPS instruction formats. In MIPS, if registers \$S0 to \$S7 map to 16 to 23 and registers \$t0 to \$t7 map on to 8 to15. Add instruction is identified by (0,32) and lw instruction is identified as 35. Convert the following MIPS assembly instruction to machine code. • Add \$t0, \$S1, \$S2 • lw \$S1,100(\$S2)	CO2	(06)
	c)	Give the IEEE 754 binary representation of the number -0.75_{10} in single precision.	CO2	(08)

UNIT - III

5.	a)	With a neat diagram describe the read and write operation on a single DRAM cell.	CO3	(10)
	b)	Consider a computer system with main memory of size 128 bytes and cache memory of size 32 bytes divided in to 8 bytes partition. Explain different cache mapping techniques to map blocks of main memory to cache lines.	CO3	(10)
6.	a)	Give the memory hierarchy of the various storage components of a computer in the order of increasing speed and cost per bit.	CO3	(05)
	b)	Describe the following terms: i. Spatial locality ii. Temporal locality iii. Cache hit iv. Cache Miss v. Write through protocol.	CO3	(10)
	c)	Consider a direct mapped cache of size 16 KB with block size 256 bytes. The size of main memory is 128 KB. Find i. Number of bits in tag ii. Tag directory size.	CO3	(05)

UNIT- IV

7.	a)	With neat diagram explain the GPU System Architecture.	CO4	(10)
	b)	List out the difference of CISC and RISC style Processors.	CO4	(06)
	c)	Describe CUDA used in GPU’s.	CO4	(04)
8.	a)	List out the key aspects of GPU programs driving GPU processor architecture and explain any one in detail.	CO4	(10)
	b)	Differentiate between multiprocessor and multicore processor. List out the benefits of multiprocessor systems over uniprocessor systems.	CO4	(10)

UNIT – V

9.	a)	Discuss the various difficulties associated with creation of parallel processing programs.	CO5	(06)
	b)	Discuss the Flynns’ classification of parallel processing systems.	CO5	(08)
	c)	What is threading? Explain the hardware multithreading in detail.	CO5	(06)
10.	a)	Discuss the Classic organization of a shared memory multiprocessor.	CO5	(06)
	b)	Illustrate along with key characteristics as how GPUs vary from CPUs.	CO5	(06)
	c)	Explain the different network topologies followed in multiprocessor systems.	CO5	(08)

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